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COMMENTARY

The potential side effects of herbal preparations in domestic animals

Roger W. Byard[1] · Ian Musgrave[1]

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Abstract
A recent series of deaths in previously healthy dogs in Victoria, Australia associated with the ingestion of raw meat contaminated by indospicine derived from native Australian plants of the *Indigofera* species draws attention to the potential that exists for herbal toxicity in domestic animals. Although the efficacy of herbal remedies generally remains unproven in domestic animals, herbal preparations are being increasingly used as supplements and treatments. Issues with incorrect ingredients, inadequate processing, faulty, incomplete or inaccurate product labelling, contamination with toxins, adulteration with undeclared pharmaceutical agents and herb-herb interactions are well recognized as causes of adverse effects in humans. However, apart from of the effects of noxious weed species, the literature on herbal toxicity in domestic animals is sparse. Thus, the forensic evaluation of cases of suspected poisoning in domestic animals should also encompass an accurate description the type and dose of any herbal preparations that may have been recently administered.

Keywords Herbal medicine · Animal death · Toxicity · Veterinary forensic pathology

Herbal preparations and alternative therapies have become increasingly popular in Western communities over the past decades with, for example, spending on complementary medicines increasing in Australia by over 100% between 1996 and 2004 [1]. Disaffection with mainstream Western medical practice, concerns over potential drug side effects and a desire for autonomy in managing health and in choosing therapeutic treatment options have resulted in herbal medicines and dietary supplements contributing to multibillion dollar industries globally. Those who take herbal medicines often claim that there are no risks of side effects due to their natural origins. Unfortunately, however it appears illogical to believe that any agent that may have a therapeutic effect will not have the potential for therapeutic side effects, and pharmacoactive herbs are no different to other treatment modalities in this respect.

Numerous publications have now highlighted health issues that may arise from the use of herbal preparations involving the inadvertent use of the wrong material, inadequate processing, faulty, incomplete or inaccurate product labelling, contamination with toxins, and adulteration with undeclared pharmaceutical agents [2–5]. All of these specific issues have resulted in significant morbidity and sometimes mortality among consumers. Other issues of concern are idiosyncratic herb-herb and herb-drug interactions [6]. These problems have affected all sections of the population including children and travelers [7, 8].

It is also recognized that herbal preparations are being increasingly used as supplements and treatments for a wide range of domestic and commercial animals ostensibly to enhance health and promote growth [9], although the efficacy of herbal remedies generally remains unproven in domestic animals [10]. Herbal remedies for animals typically utilize the same herbs that are used in human medicines [11]. This overlap is recognized by the Australian Pesticides and Veterinary Medicines Authority, the regulator of veterinary medicines in Australia, which takes guidance for the approval of veterinary herbal products from the Therapeutic Goods Administration [12].

Despite the recognition that a wide range of plants are toxic for domestic animals, such as dogs, including azaleas, buttercups, cherries, lily of the valley and rhubarb [13], little has been written on the possible side effects of herbal preparations used by owners to treat pets. Isolated reports do exist of adverse effects in animals following the ingestion of herbal preparations [14]. In one series of 47 dogs that had ingested a herbal supplement containing guarana

✉ Roger W. Byard
roger.byard@sa.gov.au

[1] Adelaide Medical School, The University of Adelaide, Adelaide, South Australia, Australia

and ma huang (Ephedra spp.), the majority of animals suffered clinical manifestations including behavioral changes, vomiting, tachycardia, hyperthermia and seizures, with death occurring in 17% [15]. Pennyroyal used to treat flea infestation contains pulegone which has reportedly caused vomiting, diarrhea and hemoptysis with death within 48 h in one medicated dog [16]. Garlic has caused damage to red blood cells in dogs [17] with resultant anemia in horses that had been fed freeze-dried preparations [18]. The American Society for the Prevention of Cruelty to Animals (ASPCA) dealt with 45 cases of accidental ingestion of medications containing Echinacea species from 1992 to 2000 that had caused vomiting and drooling [10]. Despite St. John's Wort being toxic to many animal species, medicines containing St. John's Wort are occasionally given to companion animals [11] with resultant poisoning [19].

As with herbal preparations used by humans, adverse drug herbal interactions are also likely [20] and veterinary herbal medicines are often contaminated with mycotoxins, toxic elements such as heavy metals, and pesticide residues [21]. It is very likely, however, that cases of herbal poisoning in animals are under-reported and/or misdiagnosed.

Given that most common domestic animals such as cats and dogs have standard mammalian physiology and metabolic pathways [11] it would not be unexpected that they would also be susceptible to similar side effects of herbal preparations known to cause illness in humans. They are, after all, being exposed to identical issues of contamination, adulteration, product variability and herb-herb interactions [10, 20, 21]. There may also be unique species-based aspects to side effects that have not yet been elucidated. The recent deaths of 14 dogs and hospitalization of 30 others in Victoria, Australia, following the consumption of raw meat contaminated with indospicine demonstrates this point. Found in native Australian plants of the *Indigofera* species, indospicine may result in death from liver failure in dogs due to a particular canine sensitivity [22, 23].

Thus, when a case of apparent poisoning in a domestic animal such as a dog, presents for forensic evaluation, inquiries should be made about possible exposure to herbal preparations. Failure to specifically inquire about medication with herbal products, that may be considered by owners to be intrinsically harmless, may result in delay in diagnosis and/or further exposure of this and other animals to potentially lethal material. As with medicolegal cases in humans [5] we just do not have a clear idea of how significant the effects of alternative therapies are in causing or contributing to animal illnesses.

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